

Understanding Inter-Concept Relationships in Concept-Based Models

Laura Jimena Tagle Muñoz, Qinyou Wang, Sidharth Padmanabhan, Fabian Bosshard

{jimena.tagle, qinyou.wang, sidharth.padmanabhan, fabian.bosshard}@usi.ch

Abstract

Whether concept-based models learn meaningful relationships between concepts is a crucial question recently investigated by [1]. In this project, we reproduce the main evaluations of concept bases on Colored Modified National Institute of Standards and Technology (MNIST), dSprites, Caltech-UCSD Birds-200-2011 (CUB), and CheXpert, focusing on the metrics for concept stability and faithfulness. Our results largely support the paper's qualitative conclusion that label-based and co-occurrence-based concept representations recover inter-concept structure more reliably than Concept Embedding Model (CEM) and Concept Activation Vector (CAV)-based representations. In particular, we reproduce the strong performance of the Label basis and Concept2Vec on several metrics, while CEM remains weak even on the simple Colored MNIST setting. However, our Basis-Aided Intervention for Concepts (BAIC) intervention experiments, based on the procedure described in Section 4.4 and Algorithm 1, do not reproduce the reported improvement on CUB, suggesting that intervention propagation is sensitive to the underlying model, implementation details, and concept calibration. We also identify theoretical issues in the original concentration and intervention arguments and discuss corrected formulations in Theorem 1 and Theorem 2. Overall, we find that the main claim about the difficulty of learning stable and faithful inter-concept relationships is reproducible, while the downstream intervention benefit of BAIC is less robust in our setup.

1 Introduction

Concept-based models try to make neural networks easier to interpret by expressing predictions through human-interpretable concepts. In a Concept Bottleneck Model (CBM), the simplest of the family, the model first predicts a vector of concepts and then uses this vector to predict the final label [2]. Other methods use concepts differently, for example as directions in activation space [3], as learned concept embeddings [4], or as post-hoc and label-free bottlenecks [5, 6]. This line of work is attractive because it gives users a way to inspect and sometimes correct a model in terms that are closer to the task itself.

However, concept-based explanations are not automatically reliable. Previous work has shown issues such as leakage through the concept layer [7, 8], uncertainty in human concept interventions [9], and brittle or manipulable explanations [10, 11]. [1] investigates a related but more specific question: do concept-based models actually capture relationships between concepts? This is a natural question, since human reasoning often uses such relationships [12], and datasets such as CUB contain many correlated visual attributes [13].

[1] compares several *concept bases* (see Section 4.3), including bases derived from CEMs, CAVs, Concept2Vec-style embeddings inspired by word2vec [14], and label co-occurrence. Its main empirical message is that learned concept bases are often unstable and do not reliably recover known inter-concept structure. The authors then introduce BAIC, a test-time intervention method that uses a concept basis to propagate information from corrected concepts to related uncorrected ones.

In this project, we reproduce the core claims of [1]: whether different concept bases recover meaningful inter-concept relationships, and whether BAIC improves over ordinary concept interventions.

2 Related Work

Concept-based explainability goals and weaknesses

Deep learning models are well known for achieving high predictive performance, but their internal decision-making process is difficult to interpret. This is problematic in certain domains such as medicine, fraud detection, and insurance, where model decisions may affect a person's health, finances, or legal situation. In these settings, it is not enough to know what the model predicted; it is also important to understand why the model made that prediction. Concept-based explainability aims to fill the gap between predictive accuracy and interpretability by explaining model decisions using human-understandable concepts.

Recent approaches include CBMs and CEMs, which are discussed in Section 3. These models can make predictions more interpretable, but prior work has shown that they also have important failure modes. One such issue is **concept-task leakage**, where the concept predictor learns information about the final task label instead of only detecting the intended concepts [7, 8]. For example, if an image looks like a bird, the model may predict concepts such as “wings” or “feathers” even when they are not clearly visible. Another issue is caused by **concept correlations**, where a model predicts one concept because a related concept is present, rather than because the concept itself is actually visible [1]. These weaknesses suggest that concept-based models may appear interpretable while still producing explanations that are not fully reliable.

Inter-concept relationships. Human reasoning often relies not only on individual concepts, but also on relationships between concepts [15, 16, 17]. In this paper, we observe that inter-concept relationships are mainly studied through similarities between concept representations, often reflecting whether concepts are close in representation space or frequently co-occur in the data.

Building on these ideas, the paper does not investigate whether *individual* concept predictions are correct. Instead, it asks whether concept-based models learn meaningful *relationships between concepts*.

Importantly, we noticed that the paper does not propose a new concept-based model that directly solves these weaknesses. Rather, it identifies concept representations with strong inter-concept relationships as a possible direction for improving future concept-based models.

3 Background

Concept-based models, such as CBMs and CEMs, usually follow a *two-stage prediction process*. First, a concept predictor g predicts whether a set of human-understandable concepts are present in the input x . Then, a label predictor f uses those predicted concepts $\hat{c}$ to produce the final predicted label $\hat{y}$:

$$x \xrightarrow{g} \hat{c} \xrightarrow{f} \hat{y}$$

For example, given one image x , the model may first predict concepts such as: $\hat{c} = [\text{wings} = 0.92, \text{feathers} = 0.88, \text{four legs} = 0.05]$ and then use these concept predictions to classify the image as a bird.

These models also allow *concept intervention*. At test time, an expert can correct one or more predicted concepts before the final label is computed. For instance, if the model incorrectly predicts feathers = 0, an expert can correct it to feathers = 1. The corrected concept vector is then passed to the label predictor f , potentially improving the final prediction.

4 Methods

4.1 Concepts

Concept-based learning is a supervised learning setup. In addition to input samples $X = \{\mathbf{x}^{(i)} \in \mathbb{R}^m\}_{i=1}^n$ and labels $Y = \{y^{(i)} \in \{1, \dots, L\}\}_{i=1}^n$, we are also given concept annotations $C = \{\mathbf{c}^{(i)} \in \{0, 1\}^k\}_{i=1}^n$. That is, each data point $\mathbf{x}^{(i)}$ has an associated binary vector $\mathbf{c}^{(i)}$ of k concepts, where $c_j^{(i)} = 1$ if the j^{th} concept is present and $c_j^{(i)} = 0$ otherwise.

4.2 Ground-Truth Baseline: Label Basis

The *label basis* is $\{\mathbf{v}^{(j)} \in \{0, 1\}^n\}_{j=1}^k$, where the *label vector* for the j^{th} concept is defined as $\mathbf{v}^{(j)} := [c_j^{(1)}, \dots, c_j^{(n)}]^\top \in \{0, 1\}^n$, i.e. $v_i^{(j)}$ encodes if concept j is present in the i^{th} data point.

Key Insight In the label basis, a concept is represented by **where it appears in the dataset**. Two concepts are close if they **co-occur in the same images**. ▶

Moreover, it is clear that the dot product $\mathbf{v}^{(a)} \cdot \mathbf{v}^{(b)}$ counts the number of training samples in which both concepts are active. Thus, the empirical conditional co-occurrence of concept b given concept a is

$$\hat{M}_{a,b} := \frac{\mathbf{v}^{(a)} \cdot \mathbf{v}^{(b)}}{\|\mathbf{v}^{(a)}\|_1} = \frac{\sum_{r=1}^n c_a^{(r)} c_b^{(r)}}{\sum_{r=1}^n c_a^{(r)}}$$

estimating the probability $M_{a,b} = \mathbb{P}(c_b = 1 \mid c_a = 1)$, implying that the label basis encodes inter-concept relationships directly through empirical co-occurrence: two concepts are considered related when one is likely to be active whenever the other is active.

Remark Since the label basis is constructed directly from the ground-truth concept annotations, it achieves all of the desiderata (i), (ii), (iii), (iv) from Section 4.3.1, placing an upper bound on the performance of other concept bases (see Section 4.3). ▶

Theorem 1. Assume that the concept vectors $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(n)} \in \{0, 1\}^k$ are independent samples from a common joint distribution $p_c(\cdot)$ on $\{0, 1\}^k$ (no independence between the coordinates $c_1, \dots, c_k$ is assumed). Let

$$\pi_{\min} := \min_{1 \leq a \leq k} \mathbb{P}(c_a = 1)$$

and assume that $\pi_{\min} > 0$. Then, for every $\epsilon, \delta \in (0, 1)$,

$$\mathbb{P}\left(\max_{1 \leq a, b \leq k} |\hat{M}_{a,b} - M_{a,b}| \geq \epsilon\right) \leq \delta$$

whenever $n \geq \max\left\{\frac{8}{\pi_{\min}} \log \frac{2k}{\delta}, \frac{1}{\pi_{\min} \epsilon^2} \log \frac{4k^2}{\delta}\right\}$. ◀

Proof Fix $a \in \{1, \dots, k\}$ and define $N_a := \sum_{r=1}^n c_a^{(r)}$. N_a is the number of samples in the dataset in which concept a is active. Since each $c_a^{(i)}$, $i \in \{1, \dots, n\}$, is Bernoulli with parameter $\mathbb{P}(c_a = 1)$, we have

$$\mu := \mathbb{E}[N_a] = n \mathbb{P}(c_a = 1)$$

By the multiplicative Chernoff bound for sums of independent Bernoulli random variables [18, p. 3], $\mathbb{P}(N_a \leq (1 - \eta)\mu) \leq \exp(-\frac{\eta^2 \mu}{2})$ for $0 <$

$\eta \leq 1$. Taking $\eta = 1/2$, we obtain

$$\mathbb{P}\left(N_a < \frac{1}{2}\mu\right) \leq \mathbb{P}\left(N_a \leq \frac{1}{2}\mu\right) \leq \exp\left(-\frac{\mu}{8}\right)$$

Substituting $\mu = n \mathbb{P}(c_a = 1)$, this becomes

$$\mathbb{P}\left(N_a < \frac{n \mathbb{P}(c_a = 1)}{2}\right) \leq \exp\left(-\frac{n \mathbb{P}(c_a = 1)}{8}\right)$$

Since $\mathbb{P}(c_a = 1) \geq \pi_{\min}$, we also have $\frac{n \pi_{\min}}{2} \leq \frac{n \mathbb{P}(c_a = 1)}{2}$. Hence

$$\left\{N_a < \frac{n \pi_{\min}}{2}\right\} \subseteq \left\{N_a < \frac{n \mathbb{P}(c_a = 1)}{2}\right\}$$

and therefore

$$\mathbb{P}\left(N_a < \frac{n \pi_{\min}}{2}\right) \leq \exp\left(-\frac{n \mathbb{P}(c_a = 1)}{8}\right) \leq \exp\left(-\frac{n \pi_{\min}}{8}\right)$$

If we define $E_a := \{N_a < \frac{n \pi_{\min}}{2}\}$ then the event that at least one concept appears fewer than $n \pi_{\min}/2$ times is $\bigcup_{a=1}^k E_a$ and from above we have $\mathbb{P}(E_a) \leq \exp(-\frac{n \pi_{\min}}{8})$ for all $a \in \{1, \dots, k\}$.

Now fix $a, b \in \{1, \dots, k\}$. Let $\mathcal{A}_a := \{i \in \{1, \dots, n\} : c_a^{(i)} = 1\}$ denote the index set for which concept a is active. We have $|\mathcal{A}_a| = N_a$. Note that $\mathcal{A}_a$ is a random set and $N_a = |\mathcal{A}_a|$ is a random variable, since both depend on the sampled concept annotations $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(n)}$, and therefore on which samples happen to contain concept a .

Fix a possible active set $A \subseteq \{1, \dots, n\}$ with $|A| > 0$. Conditioned on the event $\{\mathcal{A}_a = A\}$, we have

$$\hat{M}_{a,b} = \frac{1}{|A|} \sum_{i \in A} c_b^{(i)}$$

Moreover, the random variables $c_b^{(i)}$, $i \in A$, are independent, take values in $[0, 1]$, and have mean

$$\mathbb{E}[c_b^{(i)} \mid \mathcal{A}_a = A] = \mathbb{P}(c_b = 1 \mid c_a = 1) = M_{a,b}$$

Therefore, Hoeffding's inequality [19, p. 16, Thm. 2] gives

$$\mathbb{P}(|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon \mid \mathcal{A}_a = A) \leq 2 \exp(-2|A|\epsilon^2)$$

In particular, if $|A| \geq n \pi_{\min}/2$, then

$$\mathbb{P}(|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon \mid \mathcal{A}_a = A) \leq 2 \exp(-n \pi_{\min} \epsilon^2)$$

Let $G = \bigcap_{a=1}^k E_a^c$. This event says that every concept appears “often enough”, i.e. $N_a \geq n \pi_{\min}/2$ for all $a \in \{1, \dots, k\}$. From the previous paragraph, $\mathbb{P}(G^c) \leq k e^{-n \pi_{\min}/8}$. For each fixed pair (a, b) , the Hoeffding bound above implies

$$\begin{aligned} & \mathbb{P}\left(\{|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon\} \cap G\right) \\ & \leq \mathbb{P}\left(\{|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon\} \cap \{|\mathcal{A}_a| \geq n \pi_{\min}/2\}\right) \\ & = \sum_{\substack{A \subseteq \{1, \dots, n\} \\ |A| \geq n \pi_{\min}/2}} \mathbb{P}(|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon \mid \mathcal{A}_a = A) \mathbb{P}(\mathcal{A}_a = A) \\ & \leq \sum_{\substack{A \subseteq \{1, \dots, n\} \\ |A| \geq n \pi_{\min}/2}} 2 \exp(-n \pi_{\min} \epsilon^2) \mathbb{P}(\mathcal{A}_a = A) \\ & = 2 \exp(-n \pi_{\min} \epsilon^2) \sum_{\substack{A \subseteq \{1, \dots, n\} \\ |A| \geq n \pi_{\min}/2}} \mathbb{P}(\mathcal{A}_a = A) \\ & \leq 2 \exp(-n \pi_{\min} \epsilon^2) \end{aligned}$$

Therefore,

$$\begin{aligned}
& \mathbb{P} \left(\max_{1 \leq a, b \leq k} |\hat{M}_{a,b} - M_{a,b}| \geq \epsilon \right) \\
&= \mathbb{P} \left((\{\max \geq \epsilon\} \cap G^c) \cup (\{\max \geq \epsilon\} \cap G) \right) \\
&\leq \mathbb{P}(G^c) + \mathbb{P} \left(\left\{ \max_{1 \leq a, b \leq k} |\hat{M}_{a,b} - M_{a,b}| \geq \epsilon \right\} \cap G \right) \\
&= \mathbb{P} \left(\bigcup_{a=1}^k E_a \right) + \mathbb{P} \left(\bigcup_{a=1}^k \bigcup_{b=1}^k \left(\{|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon\} \cap G \right) \right) \\
&\leq \sum_{a=1}^k \mathbb{P}(E_a) + \sum_{a=1}^k \sum_{b=1}^k \mathbb{P} \left(\{|\hat{M}_{a,b} - M_{a,b}| \geq \epsilon\} \cap G \right) \\
&\leq k e^{-\frac{n\pi_{\min}}{8}} + \sum_{a=1}^k \sum_{b=1}^k 2e^{-n\pi_{\min}\epsilon^2} \\
&= k e^{-\frac{n\pi_{\min}}{8}} + 2k^2 e^{-n\pi_{\min}\epsilon^2}
\end{aligned}$$

The first term is at most $\delta/2$ whenever $n \geq 8\pi_{\min}^{-1} \log(2k/\delta)$ and the second term is at most $\delta/2$ whenever $n \geq \pi_{\min}^{-1} \epsilon^{-2} \log(4k^2/\delta)$. Hence, by taking the larger of the two, the whole probability is at most δ , as claimed. $\square$

4.3 Concept Bases

Concept-based models learn a set of concept vectors, a **concept basis** $B = \{\mathbf{v}^{(j)} \in \mathbb{R}^d\}_{j=1}^k$, where each vector $\mathbf{v}^{(j)}$ is a d -dimensional representation of concept j .

The precise extraction of these vectors depends on the model.

1. For CAVs, the concept vector is instead a direction in activation space, learned by separating examples where the concept is present from examples where it is absent [3].
2. For CEMs [4], the model produces sample-dependent concept embeddings $\mathbf{z}_j^{(i)} \in \mathbb{R}^d$ [4]. A global concept vector can be obtained by averaging over all training samples in which concept j is active,

$$\hat{\mathbf{v}}^{(j)} := \frac{1}{|\mathcal{A}_j|} \sum_{i \in \mathcal{A}_j} \mathbf{z}_j^{(i)}$$

where $\mathcal{A}_j := \{i \in \{1, \dots, n\} : c_j^{(i)} = 1\}$ denotes the index set for which concept j is present.

3. For Concept2Vec, concept vectors are learned directly from concept co-occurrences. This is done with a Skip-gram-style objective, analogous to word2vec [14]. The model is trained to distinguish pairs of concepts that occur in the same data point from pairs sampled from different data points. Thus, concepts that frequently appear together are encouraged to have similar, nearly parallel embeddings.

4.3.1 Desiderata for Concept Bases

According to [1, 20], well-calibrated concept representations should satisfy the following desiderata:

- (i) **stability**: the learned concept geometry should be largely reproducible across different random initializations (seeds)
- (ii) **robustness**: small perturbations of the input should not substantially change the inferred relationships between concepts
- (iii) **responsiveness**: meaningful changes to the input should be reflected by corresponding changes in the recovered concept relationships
- (iv) **faithfulness**: the relationships encoded by the concept representations should agree with known concept dependencies in the data

4.3.2 Distances between Concept Bases

Let δ_v be a distance metric on the space of concept vectors, e.g. the ℓ_2 -norm. For a concept vector $\mathbf{v}^{(j)} \in B$, we write $N_B(\mathbf{v}^{(j)})$ for the set of its t nearest neighbors in B , where t is a hyperparameter. The distance between two concept bases $B_1 = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(k)}\}$, $B_2 = \{\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(k)}\}$ is then defined as

$$\delta_b(B_1, B_2) := 1 - \frac{1}{k} \sum_{i=1}^k \frac{|N(\mathbf{v}^{(i)}) \cap N(\mathbf{u}^{(i)})|}{t} \quad (1)$$

i.e. one minus the average relative nearest-neighbor overlap.

Key Insight (1) measures how much the **nearest-neighbor structure** of two concept bases differs. For each concept, we compare its t nearest neighbors in both bases. If the same concepts are nearby in both bases, then the distance is small. If the neighborhoods barely overlap, then the distance is large. $\blacktriangleleft$

4.3.3 Metrics for Concept Bases

Using (1), we can now formalize the desiderata introduced in Section 4.3.1 and turn them into quantifiable metrics for concept bases:

- (i) **stability**: Let B_1 and B_2 denote concept bases extracted from independently trained models with different seeds. The stability score is defined as

$$1 - \mathbb{E}[\delta_b(B_1, B_2)]$$

where the expectation is taken over random initializations. Higher values indicate that the learned inter-concept structure is reproducible.

- (ii) **robustness**: Let B denote a concept basis extracted from an unperturbed dataset and let B' denote the basis obtained from a slightly perturbed version of the same data. Robustness is measured as

$$1 - \delta_b(B, B')$$

where larger values indicate that the neighborhood structure of concepts remains largely unchanged under small perturbations.

- (iii) **responsiveness**: Let B be a concept basis learned from a baseline dataset and let $\tilde{B}$ be a basis learned from a heavily corrupted dataset. The responsiveness score is

$$\delta_b(B, \tilde{B})$$

since, in this setting, large differences between the two bases are desirable. Whereas robustness should resist small amounts of noise, responsiveness should react to meaningful distribution changes.

- (iv) **faithfulness**: Define vectors $\{\mathbf{s}^{(1)}, \dots, \mathbf{s}^{(k)}\}$ where the l -th component of $\mathbf{s}^{(j)}$ quantifies the importance of concept j for label l . We then construct the **Shapley basis** $B_s = \{\mathbf{s}^{(1)}, \dots, \mathbf{s}^{(k)}\}$ and compare it to the learned concept basis B . Faithfulness is defined as

$$1 - \delta_b(B_s, B)$$

To compute the entries of $\mathbf{s}^{(j)}$, [1] uses a quantity inspired by Shapley values [21],

$$s_l^{(j)} := \sum_{i \in \bar{\mathcal{A}}_j} f(g(\mathbf{x}^{(i)}))_l - \sum_{i \in \bar{\mathcal{A}}_j} f(g(\mathbf{x}^{(i)}))_l$$

where $\bar{\mathcal{A}}_j := \{1, \dots, n\} \setminus \mathcal{A}_j$, g denotes the concept predictor and f the label predictor. Intuitively, this measures how strongly the presence of concept j affects predictions for label l .

Key Insight In the Shapley basis or **s-basis**, a concept is represented by **how it affects the task labels**. Two concepts are close if they **similar influence on class predictions**. $\blacktriangleleft$

4.4 Basis-Aided Concept Intervention

The goal of concept intervention is to correct some predicted concepts at test time and then recompute the task prediction. A standard intervention only changes the concepts explicitly corrected by the expert. BAIC instead uses the concept basis to also update concepts that were not directly intervened on. The underlying assumption is that nearby concepts in a good concept basis are semantically related.

Let $B = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(k)}\}$ be a concept basis, and let δ_v be the distance used to compare concept vectors. For a test sample $\mathbf{x}^{(i)}$, suppose that an expert intervenes on the concepts in $\mathcal{I} \subseteq \{1, \dots, k\}$, providing their ground-truth values $c_j^{(i)}$, $j \in \mathcal{I}$. For an unintervened concept $a \notin \mathcal{I}$, BAIC first finds the q closest concepts to a in the concept basis, then keeps only those that were intervened on, and averages their corrected concept values.

Algorithm 1 Basis-Aided Concept Intervention

Require: concept predictor g , label predictor f , concept basis $B = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(k)}\}$, test input $\mathbf{x}^{(i)}$, intervened concept set $\mathcal{I}$, ground-truth values $\{c_b^{(i)} : b \in \mathcal{I}\}$, number of neighbors q

Ensure: predicted label $\hat{y}^{(i)}$

```

1:  $\hat{\mathbf{c}}^{(i)} \leftarrow g(\mathbf{x}^{(i)})$ 
2: for  $b \in \mathcal{I}$  do
3:    $\hat{c}_b^{(i)} \leftarrow c_b^{(i)}$ 
4:  $\mathcal{U} \leftarrow \{1, \dots, k\} \setminus \mathcal{I}$  ▷ unintervened concepts
5: for  $a \in \mathcal{U}$  do
6:    $N_q(a) \leftarrow \arg \min_{b \in \{1, \dots, k\} \setminus \{a\}} \delta_v(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ 
7:    $I(a) \leftarrow N_q(a) \cap \mathcal{I}$ 
8:   if  $|I(a)| > 0$  then
9:      $\hat{c}_a^{(i)} \leftarrow \frac{1}{|I(a)|} \sum_{b \in I(a)} c_b^{(i)}$ 
10: return  $f(\hat{\mathbf{c}}^{(i)})$ 

```

Theorem 2. Let $\underline{M} \in \mathbb{R}^{k \times k}$, and suppose that $M'_{a,b} = M_{a,b} + \xi_{a,b}$, where the $\xi_{a,b}$ are independent $\mathcal{N}(0, \sigma^2)$ random variables. Let $\mathcal{I} \subseteq \{1, \dots, k\}$ be the set of intervened concepts and $\mathcal{U} = \{1, \dots, k\} \setminus \mathcal{I}$. For each $a \in \mathcal{U}$, let $I(a) \subseteq \mathcal{I}$ be the set of intervened neighbors used to update concept a , and define $\mathcal{U}_+ := \{a \in \mathcal{U} : I(a) \neq \emptyset\}$. For each $a \in \mathcal{U}_+$, let $\beta_a = \arg \max_{b \in I(a)} M_{a,b}$, $\beta'_a = \arg \max_{b \in I(a)} M'_{a,b}$. Then

$$\mathbb{E} \left[\sum_{a \in \mathcal{U}_+} (M_{a,\beta_a} - M_{a,\beta'_a}) \right] \leq \sum_{a \in \mathcal{U}_+} \sum_{b \in I(a)} \Phi \left(\frac{M_{a,b} - M_{a,\beta_a}}{\sqrt{2}\sigma} \right) (M_{a,\beta_a} - M_{a,b})$$

where Φ is the standard normal distribution function. ◁

Proof Fix $a \in \mathcal{U}_+$. Since β_a maximizes $M_{a,b}$ over $b \in I(a)$, the regret $M_{a,\beta_a} - M_{a,\beta'_a}$ is non-negative. Splitting according to the value selected by the noisy maximizer gives

$$\mathbb{E}[M_{a,\beta_a} - M_{a,\beta'_a}] = \sum_{b \in I(a)} \mathbb{P}(\beta'_a = b) (M_{a,\beta_a} - M_{a,b})$$

For $b = \beta_a$, the regret term is zero. Hence we may assume $b \neq \beta_a$. If $\beta'_a = b$, then the noisy value of b must be at least as large as the noisy value of β_a . Hence $\{\beta'_a = b\} \subseteq \{M'_{a,b} \geq M'_{a,\beta_a}\}$. By independence, $\xi_{a,b} - \xi_{a,\beta_a} \sim \mathcal{N}(0, 2\sigma^2)$. Therefore

$$\mathbb{P}(\beta'_a = b) \leq \mathbb{P}(M'_{a,b} \geq M'_{a,\beta_a}) = \mathbb{P}(\xi_{a,b} - \xi_{a,\beta_a} \geq M_{a,\beta_a} - M_{a,b})$$

$\xrightarrow{\text{symmetry of } \mathcal{N}} \Phi \left(\frac{M_{a,b} - M_{a,\beta_a}}{\sqrt{2}\sigma} \right)$

Substituting this into the above expectation gives the bound for fixed a . Summing over $a \in \mathcal{U}_+$ and using linearity of expectation gives the claim. ◻

5 Implementation

5.1 Datasets

The paper uses four datasets. Some of the links to download the external databases were available through the authors' GitHub repository. However, they did not come in the expected format and folder structure as the scripts were expecting, therefore we created new scripts for downloading each dataset. We considered important to understand the nature of each dataset, since the strength of inter-concept relationships can influence the performance of the different concept bases.

1. **Colored MNIST:** Each sample is an image of a colored hand-written digit. The concepts are ten digit concepts and ten color concepts, where each digit is paired with a specific color. For example, all images of digit 2 are assigned color 2, and all images of digit 3 are assigned color 3. The prediction label corresponds to the digit-color combination. Due to this pairing, the inter-concept relationships are strong and straightforward. Therefore, a good concept basis is expected to recover these relationships correctly.
2. **dSprites:** Each sample is a 2D image of a simple object. The concepts include shape, location, size, and orientation. In contrast to Colored MNIST, dSprites represents a setting where concepts are weakly related or mostly independent. For instance, a square-shaped object can appear in any location and have any orientation. Therefore, strong inter-concept relationships are not necessarily expected in this dataset.
3. **CUB:** Each sample is a bird image, and the concepts correspond to visual attributes such as bird size and wing color. Following the original paper, 112 attributes are used as concepts. The prediction label is the bird species. CUB is a more realistic dataset than Colored MNIST and dSprites, since it contains natural images and semantically meaningful concepts that may co-occur, such as similar colors across different bird body parts.
4. **CheXpert:** This is a medical chest radiograph dataset. The concepts correspond to medical annotations, from which 13 are selected. The prediction task is to classify whether the image corresponds to "no condition". This dataset represents another realistic dataset.

5.2 Implementation Details

5.2.1 Codebase.

We used the official [Concept-Learning](https://github.com/naveenr414/Concept-Learning) repository released with the paper: <https://github.com/naveenr414/Concept-Learning>. For reproducibility fixes and cluster execution, we also maintained a modified fork: <https://github.com/Sid-WC121/Concept-Learning>.

5.2.2 Environment and execution setting.

Although the repository provides general setup instructions, in our experience the full pipeline ran reliably only on Linux. The final experiments were executed on the USI ROSA cluster. We used the `slim` partition for Central Processing Unit (CPU)-heavy jobs and the `gpu` partition for CEM training when Graphics Processing Unit (GPU) execution was possible.

5.2.3 Study pipeline.

Our reproduction consisted of four stages:

1. **Dataset preparation.** We downloaded and prepared the four datasets using our developed scripts. In addition, we generated the perturbed dataset variants needed for the Robustness and Responsiveness metrics.

2. **Basis generation.** We generated Label, Concept2Vec, Testing with Concept Activation Vectors (TCAV), and CEM concept bases. There were found many independent functions in the repository, therefore we develop scripts for running the full pipeline. Seed 43 was used for the main metric runs, while seeds 44 and 45 were additionally generated for the Stability metric.
3. **Metric support models.** For the Faithfulness metric, we trained frozen VGG16 classifiers for each dataset. The frozen model loading script was also missing in the repository. This model is used only for evaluation.
4. **Evaluation and plotting.** We computed the final metrics and generated the report figures.

5.2.4 Hyperparameters.

We followed the default settings. In particular. For MNIST and dSprites, CEM was trained for 50 epochs with a ResNet34 backbone, using seeds {43,44,45}. The same pipeline was also used to generate the auxiliary robustness and responsiveness variants required by the metric computations. For CUB, Concept2Vec was trained with embedding size 32 for 5 epochs. TCAV was executed through separate Slurm CPU jobs. CEM was trained separately for 50 epochs, again using a ResNet34 backbone and seeds {43,44,45}.

5.2.5 Reproducibility artifacts.

The final artifacts we saved include: basis vectors for CEM, TCAV, and Concept2Vec; CEM checkpoints and configuration files; frozen VGG16 classifiers for the Faithfulness metric; metric outputs and generated figures. The repository did not provide ready-to-use saved outputs for our setup, so these artifacts had to be generated from scratch, except for pretrained backbone models downloaded for TCAV.

5.3 Compute Budget and Logging

5.3.1 Runtime.

GPU jobs ran on a single NVIDIA GeForce GTX 1080, 1080 Ti, or A100 PCIe 40GB, typically with 4 CPU cores and 32 GB of RAM. CPU-only runs used the `slim` partition with 2–4 cores and 16 GB of RAM. Successful basis-training required 18.5 hours in total, with robustness and responsiveness generation requiring a similar duration. Support jobs, such as computing truthfulness metrics with a VGG16 classifier, took 12.5 hours, while CUB intervention experiments took 10.8 hours on the `slim` partition.

5.3.2 Engineering fixes and debugging.

Significant effort was required to adapt the repository to run reliably:

- Writing scripts to link unconnected functions
- Fixing internal inconsistencies (e.g., CUB assuming 312 concepts in some places and 250/112 in others)
- Correcting missing paths and machine-specific prefixes
- Fixing typos that broke function calls
- Enabling GPU training for CUB
- Removing unused packages
- Fixing CEM checkpoint naming to prevent overwriting across datasets

5.3.3 Reproducibility note.

Because the original repository is not push-button runnable, our reproduction combines the official codebase with the minor engineering fixes required to execute the pipeline.

6 Results

6.1 Table 1: Concept Bases on MNIST

For Table 1 of [1], our reproduction matches the qualitative conclusion of the paper. TCAV, Concept2Vec, and the Label basis achieve perfect

Basis	Concept Agreement ($\uparrow$)
TCAV	100% $\pm$ 0%
CEM	8% $\pm$ 2%
Concept2Vec	100% $\pm$ 0%
Label	100% $\pm$ 0%

Table 1. Concept agreement on Colored MNIST. Higher is better. In Colored MNIST, the correct neighbor of each color concept is the corresponding digit concept, and vice versa (for example, `3_color` should be closest to `3_number`). The score is the fraction of concepts for which this nearest-neighbor pairing is correct, averaged across seeds.

agreement, while CEM remains clearly worse. Although our exact CEM agreement value does not match the reported number exactly, it still supports the same point: CEM does not preserve the intended color-digit pairing as reliably as the other bases, even in MNIST, where the concept relationships are simple and intuitive.

CEM results reported in the table correspond to the default CEM training on Colored MNIST without imbalance configuration, since the paper does not document this imbalance-weighted MNIST setting. We also performed an additional training with enabled imbalanced weighting, resulting on a CEM concept agreement of 5% $\pm$ 4%. A likely reason for the mismatch in the exact CEM value is that this metric is sensitive to small changes in the learned embedding geometry. For this reason, we interpret the MNIST result primarily as evidence of the relative weakness of CEM on inter-concept structure.

6.2 Figure 2: Evaluation metrics

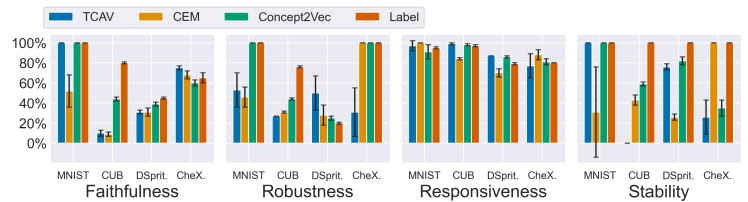


Figure 1. 4-Metric evaluation of different concept bases on four datasets

The Label Basis is the ground-truth baseline. In Figure 1 of [1], as well as in our results showcased in Figure 1, Label achieves the highest performance in (almost) all metrics and datasets. As expected, the dataset in which Label did not perform as well is dSprites, known for having weaker inter-concept relationships.

Concept2Vec is the second best, which is a basis similar in spirit to Label.

On the other hand, CEM performed poorly even in the most "intuitive-related" MNIST dataset. Both TCAV and CEM failed under realistic challenges such as CUB and CheXpert.

6.3 Figure 3: Semantic similarity between concepts

To reproduce the dendrograms of Figure 3 of [1], we generated the dendrograms from the learned concept bases using the base seed 43, matching the single-seed qualitative setup used in the repository notebooks. Concretely, for each method we loaded the concept vectors for all 112 attributes, computed pairwise concept distances, and applied Ward hierarchical clustering to obtain the tree structure. Since the repository did not provide a standalone command-line script for this exact CUB workflow, we created a script to assemble the existing hierarchy code and save the resulting dendrogram figures.

For Concept2Vec on CUB, we observed meaningful semantic grouping. In the left two dendrograms in Figure 2, concepts associated with the color *gray* cluster together, and within that cluster, *gray upper tail*

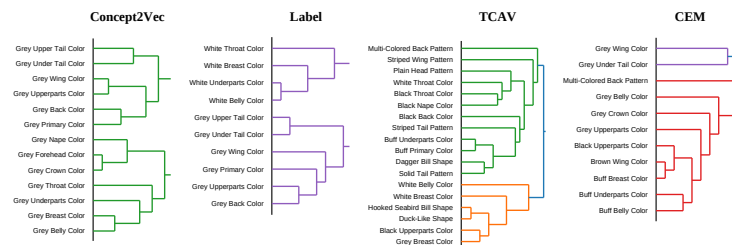


Figure 2. Dendrograms to assess semantic similarity among concepts.

and *gray under tail* appear as particularly similar concepts. The Label basis shows the same qualitative behavior and is consistent with the paper: semantically related concepts such as *white underpart* and *white belly*, as well as *gray upper tail* and *gray under tail*, are grouped together.

By contrast, the right two dendrograms in Figure 2 show that the local groupings produced by TCAV and CEM do not display the same semantic coherence. The concepts clustered together in the reported portions of these dendrograms do not exhibit an obvious shared meaning, which is consistent with the qualitative claim of the paper that TCAV and CEM fail to recover a clear semantic structure.

6.4 Figure 6: Concept Intervention on CUB

To reproduce Figure 6 of [1], we implemented our own standalone intervention workflow for CUB, based on the intervention logic scattered across the released repository. In summary, it loads a pretrained model, selects a fraction of concept groups to intervene on, propagates interventions through the chosen basis when applicable, and measures the resulting downstream classification accuracy. Although this implementation follows the intervention procedure described in the paper as closely as possible, we were not able to reproduce the same qualitative result. This suggests that the effectiveness of the label basis depends strongly on the quality and calibration of the underlying CEM model: if the learned concept representation is not sufficiently aligned with the intended semantic structure, propagating interventions can hurt the downstream performance, even with the label basis.

7 Theoretical Extension

From Inter-Concept Relationships to Tensorial Coupling and Semantic Synchronization

Although this project is primarily a reproducibility study, the reproduced work also motivates a broader theoretical question: what does it mean for a concept-based model to learn relationships between concepts? The original paper studies concept representations in order to understand whether concept-based models capture inter-concept relationships. It also shows that such relationships can be useful for downstream applications, especially concept intervention. In particular, the paper introduces Basis-Aided Concept Intervention (BAIC), where a concept basis is used to impute unintervened concepts from related intervened concepts.

The purpose of this theoretical extension is not to modify the empirical reproduction results, but to clarify the semantic and mathematical assumption that makes BAIC meaningful. BAIC is not only a test-time intervention method; it is also based on a deeper relational intuition. If a concept intervention is useful only for the directly corrected concept, then concepts are treated as independent units. If, however, correcting one concept can help correct or impute other related concepts, then the model must contain some form of relational organization among concepts. BAIC is precisely an algorithmic use of this structure. In this sense, BAIC can be interpreted as a method that operates not only on

concept values, but also on the geometry and dependency structure of the concept space.

This assumption has both an interpretability meaning and a philosophical meaning. From the interpretability perspective, a concept-based model is useful only if its concepts correspond to meaningful human-understandable properties. However, human understanding rarely operates on isolated concepts. When humans reason about an object, they do not only identify independent attributes; they also use relations among attributes. For example, in a bird image, wing color, body color, beak shape, size, and species are not interpreted as completely separate facts. They form a structured semantic field. Some attributes support, constrain, or disambiguate the interpretation of others. This is precisely the intuition that makes BAIC meaningful: intervening on one concept may also provide information about other related concepts.

From a philosophical perspective, this suggests that meaning is relational rather than purely atomistic. A concept does not acquire its full meaning only from its isolated definition, but also from its position in a network of other concepts, contexts, tasks, and possible actions. This view echoes Wittgenstein’s statement in the *Tractatus Logico-Philosophicus*: “the limits of my language mean the limits of my world” (*Die Grenzen meiner Sprache bedeuten die Grenzen meiner Welt*) [22, Prop. 5.6]. In the context of concept-based models, the available concepts determine the boundary of the model’s explicit semantic world, while the relationships among these concepts determine the internal structure of that world. If a model has only isolated concept names but no reliable relationships among them, its semantic world is fragmented—a vocabulary without grammar. In contrast, a stable and faithful concept basis gives the model a coherent semantic space where interventions can propagate meaningfully. Thus, BAIC moves beyond the vocabulary level: it does not only ask whether a model has concepts, but whether it can use relations between concepts to make interventions more effective, reflecting the general principle that meaningful explanation requires structured relations among symbols, not just the symbols themselves.

This point is important for interpretability. A model may appear interpretable because it exposes a list of human-readable concepts, but this does not guarantee that the model organizes those concepts in a human-meaningful way. A vocabulary without grammar is not yet a language; similarly, a list of concepts without reliable inter-concept relations is not yet a coherent semantic system. BAIC is interesting precisely because it moves beyond the vocabulary level. It does not only ask whether a model has concepts, but also whether the model can use the relation between concepts to make interventions more effective. Thus, BAIC can be seen as a small technical instance of a more general principle: meaningful cognition and meaningful explanation require not only symbols, but also structured relations among symbols.

This philosophical point can also be illustrated by biological and developmental examples. In honey bees, the waggle dance is a communication system used to transmit information about the direction, distance, and quality of food sources. Although this behavior has an innate biological basis, recent work suggests that accurate waggle dancing is also shaped by social learning: young bees that lack the opportunity to observe experienced dancers produce dances with larger errors, especially in distance encoding. Thus, even in a non-human communication system, meaningful signalling is not only a fixed individual capacity, but also depends on participation in a socially structured communication practice. In other words, the signal becomes reliable not simply because the individual has an internal mechanism, but because this mechanism is calibrated inside a community of signal users.

Human language development provides an even stronger example. Cases of severe social isolation and language deprivation, such as Genie, have been central in debates about the critical or sensitive period for first language acquisition. Genie was discovered after many years

of extreme isolation and language deprivation; after rescue, she made progress in non-verbal communication and learned some vocabulary, but she did not fully acquire normal grammatical language. Historical cases such as Victor of Aveyron have also been discussed in relation to delayed language acquisition and social deprivation. These cases must be interpreted carefully, because they involve trauma, deprivation, and many confounding factors. Nevertheless, they support the idea that language is not merely an optional external tool added to an already complete intelligence. Rather, language and social interaction appear to participate in the formation of structured cognition, social understanding, and the ability to organize experience conceptually.

These examples are relevant to BAIC because concept-based models also depend on an explicit semantic structure. A concept basis is not just a list of labels, in the same way that a language is not just a list of words. What matters is whether the system has access to stable relations among concepts. BAIC operates precisely on this relational level: it assumes that intervened concepts can inform the state of related uninvolved concepts. Therefore, the biological and developmental examples above motivate the same general point as BAIC: meaningful cognition and meaningful intervention require not only individual symbols, but also a structured system of relations in which those symbols acquire their role.

Mathematically, the concept basis is usually written as a set of concept vectors, $B = \{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(k)}\}$. The original paper defines such a concept basis as a collection of vectors, where each vector is a representation of one concept. These vectors are not necessarily independent; instead, they define a structured set of concepts. A simple way to extract inter-concept relationships from this basis is to define a pairwise similarity matrix

$$S_{ij} = \text{sim}(\mathbf{v}^{(i)}, \mathbf{v}^{(j)}),$$

or equivalently a distance matrix

$$s_{ij} = \delta_v(\mathbf{v}^{(i)}, \mathbf{v}^{(j)}).$$

In this formulation, the concept basis induces a graph over concepts: each concept is a node, and the similarity or distance values define weighted edges between concepts. BAIC can then be interpreted as an intervention procedure on this concept graph. Correcting one node changes its state, and the learned graph determines which neighboring nodes should also be used to update or impute other concepts.

In the label basis, such a relation can be estimated through empirical co-occurrence: $\widehat{M}_{a,b} \approx P(c_b = 1 \mid c_a = 1)$, see Section 4.3. This expression shows that concept relationships can be interpreted probabilistically: if concept a is present, how likely is concept b to also be present? In the original paper, this idea is used to justify the label basis as a ground-truth baseline: concepts that frequently co-occur have close label vectors. Therefore, the label basis directly encodes inter-concept relationships through empirical co-occurrence. BAIC uses the same general principle operationally: an intervened concept can help predict an uninvolved concept when the concept basis says that these concepts are related.

However, pairwise similarity is only a first approximation. Many semantic relationships are not purely pairwise; the relation between two concepts may depend on additional contextual modes. A natural theoretical extension is therefore to replace the pairwise relation matrix S_{ij} with a higher-order tensor $T_{ij}^{\alpha\beta}$ where i, j index concepts and α, β index additional modes such as representation layers, semantic contexts, task labels, or intervention states. The entry $T_{ij}^{\alpha\beta}$ can be interpreted as the strength with which concept j in mode β influences concept i in mode α . This transition is important because it changes the interpretation of a concept basis: while a matrix mainly answers *which* concepts are close to each other, a tensor answers *under which condition* one concept influences another. Because the same

concept pair may be strongly related in one context but weakly related in another, a tensorial extension would allow BAIC's intervention mechanism to become context-sensitive rather than globally fixed.

The transition from S_{ij} to $T_{ij}^{\alpha\beta}$ is important because it changes the interpretation of a concept basis. A matrix-based concept basis mainly answers the question: which concepts are close to each other? A tensor-based concept basis can answer a richer question: under which context, layer, label, or intervention condition does one concept influence another? This is closer to how semantic reasoning works. The same visual attribute may have different relevance depending on the predicted class, and the same concept pair may be strongly related in one context but weakly related in another. Therefore, a tensorial extension of BAIC would allow the intervention mechanism to become context-sensitive rather than globally fixed.

This is particularly relevant for datasets such as CUB, where visual attributes are semantically structured. A bird species may be characterized not by a single attribute, but by a configuration of attributes. For instance, wing color, belly color, beak shape, tail pattern, and size may jointly support a species prediction. A pairwise nearest-neighbor relation may capture some of these dependencies, but a tensorial representation could in principle capture richer multi-way interactions. In this sense, BAIC can be viewed as using a static and finite approximation of a more general tensorial semantic structure.

This perspective also suggests a dynamic interpretation of BAIC. Let $x_i^{(\alpha)}(t)$ denote the state of concept i in mode α at time or intervention step t . This state may represent a concept probability, a concept embedding, or a corrected concept value. A general coupled concept dynamics can be written as

$$\dot{x}_i^{(\alpha)} = F^{(\alpha)}(x_i^{(\alpha)}) + \sum_{\beta, j} T_{ij}^{\alpha\beta} G^{\alpha\beta} \left(x_j^{(\beta)} - x_i^{(\alpha)} \right).$$

Here, $F^{(\alpha)}$ describes the internal evolution or persistence of the concept state, $G^{\alpha\beta}$ describes how information is transmitted between concept states, and $T_{ij}^{\alpha\beta}$ determines the strength of the coupling. BAIC can then be interpreted as a discrete test-time approximation of this process: when an expert intervenes on some concepts, the intervened concept values are used to update related uninvolved concepts through the learned concept basis.

To match the original BAIC algorithm more exactly, it is useful to rewrite the general dynamics in a message-passing form. Let $\mathcal{I}$ denote the set of intervened concepts. Consider the general tensorial semantic message-passing update

$$x_j^{(\beta)} \leftarrow \sum_{\alpha} \sum_{i \in \mathcal{I}} T_{ji}^{\beta\alpha} G^{\beta\alpha} (x_i^{(\alpha)}),$$

where $T_{ji}^{\beta\alpha}$ is a coupling tensor and $G^{\beta\alpha}$ is a message function. This form says that the state of an uninvolved concept j can be imputed by aggregating messages from intervened concepts i , with the strength of each message determined by the coupling structure T . The original BAIC algorithm is recovered by choosing a very specific coupling structure and a very simple message function.

Proposition 3. BAIC as a special case of tensorial semantic coupling. The original BAIC algorithm predicts an uninvolved concept j by finding the set $I(j)$ of the q most similar intervened concepts in the concept basis and setting

$$\hat{c}_j^{(i)} = \frac{1}{|I(j)|} \sum_{j' \in I(j)} c_{j'}^{(i)}.$$

If no such intervened neighbor exists, BAIC falls back to the original concept predictor, $\hat{c}_j^{(i)} = g(x^{(i)})_j$. This BAIC update is a one-step, static, hard-nearest-neighbor, pairwise special case of the tensorial semantic message-passing framework. $\triangleleft$

Proof Start from the general tensorial semantic message-passing update

$$x_j^{(\beta)} \leftarrow \sum_{\alpha} \sum_{i \in \mathcal{I}} T_{ji}^{\beta\alpha} G^{\beta\alpha}(x_i^{(\alpha)}).$$

Now impose the following simplifying assumptions.

First, restrict the system to a single representation mode:

$$\alpha = \beta = 1.$$

Then $x_j^{(\beta)} = x_j$, $x_i^{(\alpha)} = x_i$, and the tensor $T_{ji}^{\beta\alpha}$ reduces to a pairwise coupling matrix T_{ji}^{11} .

Second, choose the message function to be the identity:

$$G(x) = x.$$

Thus, an intervened concept sends its observed ground-truth concept value as the message.

Third, define the coupling weights through BAIC's hard nearest-neighbor rule. For each unintervened concept j , let $I(j)$ be the set of the q most similar intervened concepts according to the concept-basis distance $\delta_v(v^{(j)}, v^{(j')})$. Define

$$T_{jj'} = \begin{cases} \frac{1}{|I(j)|}, & j' \in I(j), \\ 0, & j' \notin I(j). \end{cases}$$

Substituting these assumptions into the general update gives

$$x_j \leftarrow \sum_{j' \in \mathcal{I}} T_{jj'} x_{j'} = \sum_{j' \in I(j)} \frac{1}{|I(j)|} c_{j'}^{(i)} = \frac{1}{|I(j)|} \sum_{j' \in I(j)} c_{j'}^{(i)}.$$

This is exactly the BAIC update rule for an unintervened concept j . If $|I(j)| = 0$, the tensorial message-passing update receives no incoming intervened-neighbor messages; the original BAIC algorithm then uses the fallback prediction $\hat{c}_j^{(i)} = g(x^{(i)})_j$. Therefore, BAIC is recovered as a one-step, static, hard-nearest-neighbor, pairwise instance of the proposed tensorial semantic coupling framework. $\square$

Proposition 3 clarifies the theoretical contribution of the extension. The claim is not that the proposed framework changes the empirical results of the paper, nor that it proves BAIC's performance advantage in all settings. Rather, the contribution is conceptual and mathematical: the original BAIC algorithm can be reinterpreted as a special case of a broader tensorial semantic coupling framework. BAIC uses hard nearest-neighbor averaging over intervened concepts; the proposed framework generalizes this rule by allowing soft coupling weights, higher-order context-dependent tensors, non-linear message functions, and iterative updates.

This dynamic interpretation clarifies the difference between independent intervention and relational intervention. In an independent intervention model, a correction affects only the concept directly modified by the expert: $x_i \leftarrow x_i^{\text{corrected}}$. In a relational intervention model such as BAIC, the correction or known concept value is used to update other concepts:

$$x_j \leftarrow \sum_{j' \in I(j)} \frac{1}{|I(j)|} c_{j'}^{(i)}.$$

A tensorial extension replaces the hard average by context-sensitive coupling:

$$x_j^{(\beta)} \leftarrow \sum_{\alpha} \sum_{i \in \mathcal{I}} T_{ji}^{\beta\alpha} G^{\beta\alpha}(x_i^{(\alpha)}).$$

This formulation makes explicit how BAIC could be generalized from static nearest-neighbor propagation to dynamic and context-sensitive concept propagation.

The language of synchronization provides another way to understand this process. In coupled dynamical systems, synchronization does

not necessarily mean that all units become identical. It means that related units become coordinated according to the structure of their coupling. Applied to concept-based models, semantic synchronization would mean that related concepts become mutually consistent after intervention. For example, concepts that describe related visual attributes may become aligned, while unrelated concepts should remain largely unaffected. Therefore, the goal is not global synchronization of all concepts, but structured partial synchronization among semantically related concept groups.

This idea can be expressed more concretely through the distinction between global and local synchronization. Global synchronization would force all concept states to converge toward the same pattern, which would be undesirable because different concepts should remain distinct. Local semantic synchronization, in contrast, only coordinates concepts that are connected by meaningful relations. For BAIC, this is exactly the desired behavior. If an intervened concept is strongly related to a group of other concepts, then those concepts should be updated consistently. If another concept is semantically unrelated, then it should not be strongly affected. Therefore, BAIC can be interpreted as seeking a controlled, local, and semantically constrained form of synchronization.

This interpretation also explains the limitations found in our reproduction. A poor concept basis does not merely reduce static representation quality; it degrades the estimated coupling structure T , causing interventions to propagate along misleading paths or miss genuine concept relations. Thus, the problem of learning a good concept basis is fundamentally the problem of estimating the correct semantic coupling geometry. The empirical metrics used in this work, stability, robustness, responsiveness, and faithfulness, are ultimately tests of whether this estimated coupling structure is reliable enough to support meaningful intervention.

Finally, this extension connects concept-based interpretability to broader questions about semantic systems and language-based artificial intelligence. A sufficiently expressive language system can be viewed as a high-dimensional network of relations among concepts, contexts, agents, and actions. If a model internalizes such a structure, its behavior may depend not only on individual concepts but also on the organization, propagation, and stabilization of relations between concepts. In a more speculative direction, this may also be relevant to questions about whether high-level cognitive properties, including aspects of consciousness, could emerge from the internalization of sufficiently rich and stable semantic coupling structures.

For BAIC specifically, the theoretical implication is clear, its effectiveness depends on whether the concept basis approximates a meaningful coupling structure. The natural future direction suggested by this view is to move beyond static pairwise nearest-neighbor propagation toward dynamic, tensorial, and context-sensitive models of concept interaction. Such models could preserve the interpretability of concept-based learning while providing a richer mathematical account of how concepts mutually constrain and stabilize one another.

List of Acronyms

BAIC	Basis-Aided Intervention for Concepts
CAV	Concept Activation Vector
CBM	Concept Bottleneck Model
CEM	Concept Embedding Model
CPU	Central Processing Unit
CUB	Caltech-UCSD Birds-200-2011
GPU	Graphics Processing Unit
MNIST	Modified National Institute of Standards and Technology
TCAV	Testing with Concept Activation Vectors

References

- [1] Naveen Raman, Mateo Espinosa Zarlenga, and Mateja Jamnik. Understanding inter-concept relationships in concept-based models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024. URL <https://proceedings.mlr.press/v235/raman24a/raman24a.pdf>.
- [2] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept bottleneck models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 5338–5348. PMLR, 2020. URL <https://proceedings.mlr.press/v119/koh20a/koh20a.pdf>.
- [3] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viégas, et al. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors. In *Proceedings of the 35th International Conference on Machine Learning*, *Proceedings of Machine Learning Research*, pages 2668–2677. PMLR, 2018. URL <https://proceedings.mlr.press/v80/kim18d/kim18d.pdf>.
- [4] Mateo Espinosa Zarlenga, Pietro Barbiero, Gabriele Ciravegna, Giuseppe Marra, Francesco Giannini, Michelangelo Diligenti, Zohreh Shams, Frederic Precioso, Stefano Melacchi, Adrian Weller, et al. Concept embedding models. In *Advances in Neural Information Processing Systems*, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/867c06823281e506e8059f5c13a57f75-Paper-Conference.pdf.
- [5] Mert Yüsekşgönül, Maggie Wang, and James Zou. Post-hoc concept bottleneck models. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/pdf?id=nA5AZ8CEyow>.
- [6] Tuomas Oikarinen, Subhro Das, Lam M. Nguyen, and Tsui-Wei Weng. Label-free concept bottleneck models. *arXiv preprint arXiv:2304.06129*, 2023. URL <https://arxiv.org/pdf/2304.06129>.
- [7] Marton Havasi, Sonali Parbhoo, and Finale Doshi-Velez. Addressing leakage in concept bottleneck models. In *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/pdf?id=tglndfn9>.
- [8] Anita Mahinpei, Justin Clark, Isaac Lage, Finale Doshi-Velez, and Weiwei Pan. Promises and pitfalls of black-box concept learning models. *arXiv preprint arXiv:2106.13314*, 2021. URL <https://arxiv.org/pdf/2106.13314>.
- [9] Katherine M. Collins, Matthew Barker, Mateo Espinosa Zarlenga, Naveen Raman, Umang Bhatt, Mateja Jamnik, Ilia Sucholutsky, Adrian Weller, and Krishnamurthy Dvijotham. Human uncertainty in concept-based ai systems. In *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*, pages 869–889. ACM, 2023. URL <https://arxiv.org/pdf/2303.12872>.
- [10] Davis Brown and Henry Kvinge. Brittle interpretations: The vulnerability of tcav and other concept-based explainability tools to adversarial attack. *arXiv preprint arXiv:2110.07120*, 2021. URL <https://openreview.net/pdf?id=a3hQPNqIFk6>.
- [11] Ann-Kathrin Dombrowski, Maximilian Alber, Christopher Anders, Marcel Ackermann, Klaus-Robert Müller, and Pan Kessel. Explanations can be manipulated and geometry is to blame. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://papers.neurips.cc/paper/9511-explanations-can-be-manipulated-and-geometry-is-to-blame.pdf>.
- [12] James L. McClelland and Timothy T. Rogers. The parallel distributed processing approach to semantic cognition. *Nature Reviews Neuroscience*, 4(4):310–322, 2003. URL https://tallinzen.net/media/readings/mcclelland_rogers_2003.pdf.
- [13] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The caltech-ucsd birds-200-2011 dataset. Technical report, California Institute of Technology, 2011. URL <https://gwern.net/doc/ai/dataset/2011-wah.pdf>.
- [14] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013. URL <https://arxiv.org/pdf/1301.3781>.
- [15] Nick Chater, Mike Oaksford, Ulrike Hahn, and Evan Heit. Bayesian models of cognition. *Wiley Interdisciplinary Reviews: Cognitive Science*, 1(6):811–823, 2010. doi: 10.1002/wcs.79.
- [16] Thomas L. Griffiths, Kevin R. Canini, Adam N. Sanborn, and Daniel J. Navarro. Unifying rational models of categorization via the hierarchical dirichlet process. In *Proceedings of the 29th Annual Conference of the Cognitive Science Society*, pages 323–328, 2007. URL https://papers.djnavarro.net/2007_hdp.pdf.
- [17] Jiayuan Mao, Chuang Gan, Pushmeet Kohli, Joshua B. Tenenbaum, and Ji-ajun Wu. The neuro-symbolic concept learner: Interpreting scenes, words, and sentences from natural supervision. *arXiv preprint arXiv:1904.12584*, 2019. URL <https://arxiv.org/pdf/1904.12584>.
- [18] Friedrich Eisenbrand. Randomized algorithms, lecture 2. Lecture notes, EPFL, 2012. URL <https://www.epfl.ch/labs/disopt/wp-content/uploads/2018/09/lec2.pdf>. Scribe: Christos Kalaitzis.
- [19] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830. URL <https://www.tandfonline.com/doi/pdf/10.1080/01621459.1963.10500830>.
- [20] Anna Hedström, Leander Weber, Dilyara Bareeva, Felix Motzkus, Wojciech Samek, Sebastian Lapuschkin, and Marina M.-C. Höhne. Quantus: An explainable ai toolkit for responsible evaluation of neural network explanations. In *Journal of Machine Learning Research*, 2023. URL <https://www.jmlr.org/papers/volume24/22-0142/22-0142.pdf>.
- [21] Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games II*, volume 28 of *Annals of Mathematics Studies*, pages 307–317. Princeton University Press, 1953. URL <https://www.rand.org/content/dam/rand/pubs/papers/2021/P295.pdf>.
- [22] Ludwig Wittgenstein. *Tractatus Logico-Philosophicus*, volume 14. Verlag von Wilhelm Engelmann, Leipzig, 1921. URL <https://people.umass.edu/klement/tlp/tlp-ebook.pdf>.